

PA4-2026

CS300 - CSC13002 - Introduction to Software Engineering

Project Assignment 4 (PA4-2026)

Total scores: 75 points

Duration: 2-3 weeks

This is a teamwork assignment.

A - Revised Use-Case Specification - 2nd submission (5 points)

Based on the TA feedback from PA3-2026, revise your use-case model and use-case specifications. Address the identified issues, provide additional details, and ensure consistency with the vision document and your implemented features.

Your team must include a `Changes.md` file that clearly lists all changes made compared to the PA3-2026 version. Use a single `Changes.md` file per PA submission, with separate sections for each revised document. Without this file, you will not receive points for this section.

B - Software Architecture: System Context Diagram (15 points)

Formally document the software architecture of your entire project, covering all features from PA1 through PA4.

Requirements:

- **Tech Stack:** clearly list and describe the technologies used for frontend, backend, database, and any other components (e.g., authentication services, cloud storage, APIs).
- **C4 Model - Level 1 (System Context Diagram):** draw and explain a System Context diagram following the C4 Model. This diagram shows your system

as a whole, the users (actors) who interact with it, and any external systems it depends on. Include a written explanation of the diagram.

Write the architecture document in Markdown format with diagrams in Mermaid syntax.

C - Software Architecture: Container Diagram and Component Diagram (20 points)

C4 Model - Level 2 (Container Diagram): draw and explain a Container diagram following the C4 Model. This diagram zooms into your system and shows the major containers (e.g., web app, mobile app, API server, database, etc.), the technologies used by each container, and how the containers communicate with each other.

C4 Model - Level 3 (Component Diagram): draw Component diagrams following the C4 Model for both the frontend and backend containers. Each diagram zooms into a container to show the major components inside it, their responsibilities, and the interactions between them. You are not required to draw Level 3 diagrams for every feature; instead, focus on the most important features that best represent the internal structure of each container.

Requirements:

- All diagrams must be drawn using **Mermaid** format. You may use standard Mermaid flowchart/graph syntax with C4-style labeling (Mermaid's experimental C4 syntax is not required).
- For each container, provide a written description of:
 - Its responsibility and the services it provides.
 - The technology/framework used.
 - How it communicates with other containers (e.g., HTTP/HTTPS, WebSocket, database connection).
- For each component in the Level 3 diagram, describe its responsibility and its relationships with other components.

Warning: The architecture diagrams must accurately reflect your actual implementation at the time of submission. Any inconsistency between the documented architecture and the source code will result in a grade penalty.

As you continue implementing features, you must keep the architecture document up to date.

D - Deployment Diagram (5 points)

Describe how your system is deployed by mapping the containers defined in Section C to the infrastructure that runs them.

Requirements:

- Draw a deployment diagram using **Mermaid** format showing the physical or cloud infrastructure nodes (e.g., web server, application server, database server, mobile device, cloud services) and the containers deployed on each node. If your system runs entirely on a local machine, treat each container as if it were deployed on a separate logical node.
- For each node, briefly describe:
 - The hardware or cloud service used (e.g., AWS EC2, Vercel, Firebase, local machine).
 - The containers or components running on that node.
 - The communication protocols between nodes (e.g., HTTPS, TCP, WebSocket).

E - Implement 2 Functional Groups using Spec Kit (25 points)

Continue implementing **2 more functional groups** from your project, covering the full stack: frontend, backend, and database.

Requirements:

- Use **Spec Kit** to drive the implementation process, following the complete spec-driven workflow.
- Each functional group must be implemented end-to-end (UI + API/logic + data persistence).
- At this stage, Spec Kit will also generate test cases for your features. You should include the generated tests in your submission; however, you are NOT yet required to fully understand or refine the test cases. Test review and refinement will be covered in PA5-2026.
- Submit:

- A video demo with narration explaining each implemented feature. Upload the video to YouTube (Unlisted or Public) and include the link in your submission.
- The complete source code (exclude `node_modules`, `venv`, build artifacts, and other generated directories).
- The Spec Kit artifacts generated during the process (specs, plans, tasks files).

F - AI Usage Report and Weekly Report (5 points)

AI Usage Report

Each team must follow the AI usage guidelines to declare whether AI tools were used during this sprint. If AI tools were used, a detailed log of all usages must be submitted.

Weekly Report

Conduct and document the following meetings: Sprint Planning, Scrum meetings, and Sprint Review. Follow the instructions in the "WeeklyReports" (provided separately) document.

Include Jira task screenshots showing task assignments and progress for this sprint.

Submission Guidelines

- All documents must be written in **English** using **Markdown** format. Diagrams should be drawn using **Mermaid** syntax.
- In each report/document section, the team must indicate who performed the work, who reviewed it, and who edited the section. Place this information on the first line immediately after the section header.
- Submit both the **original Markdown files (.md)** and **PDF files converted from Markdown**.
- Provide a **Git log** (screenshot or exported log) showing commit history of your repository.
- Compress all files into a directory named `PA4-Group[GroupId]` and submit as a zip file.

Example: `PA4-Group01.zip`

Grading Criteria

The assignment will be graded according to how well your team prepares the required documents and implementation. Criteria include:

- English writing: easy to understand, few errors.
- Well-formatted texts, diagrams, and charts.
- Changes are clearly documented in `Changes.md`.
- The revised use-case specification addresses TA feedback and is more complete than the previous version.
- The software architecture clearly describes the tech stack and includes a correct C4 Level 1 (System Context) diagram with explanation.
- The C4 Level 2 (Container Diagram) and Level 3 (Component Diagram) are drawn in Mermaid, with clear descriptions for each container and component.
- The deployment diagram correctly maps containers to infrastructure nodes with appropriate descriptions.
- The 2 implemented functional groups are functional end-to-end and follow the Spec Kit workflow.
- All architecture diagrams are consistent with the actual source code implementation. Inconsistencies will result in grade penalties.